

I've been researching Core Animation framework for the past few months. I've started with several books on the subject, but I found watching related WWDC videos most rewarding. The presenters put the content in a relevant context which makes it easier to apprehend and learn from it.

1. Introduction

One WWDC session particularly intrigued me: "2011–421 Core Animation Essentials". They presented their demo named: "Layers in Perspective", and it showed six layers, forming a flattened cube: *[Image: /images/cacube/02_cube_5sides.png]*

The sixth layer is hiding behind the layer number 5. It has a lower `zPosition` than the layer above it.

[Image: /images/cacube/03_cube_6sides.png]

They have also demonstrated opening and closing the cube formation.

[Image: /images/cacube/04_cube_filmstrip.png]

So, I have decided to demonstrate that.

Here's a link to a GitHub repo with the source code: [Core Animation 3D Cube](#)

Here's the animation:

[Image: /images/cacube/05_CoreAnimation_3D_Cube.gif]

You can also see it on [You Tube](#)

Layers in Core Animation live in 3D geometry. But a layer is a 2D construct, so Core Animation coordinate space is called a 2.5D geometry.

To illustrate that just see what happens when you mess up your transformations.

[Image: /images/cacube/06_Intersecting_Layers.png]

Layers are 2D objects, they don't understand the third dimension. They are like playing cards in space., and there is no depth buffer available.

And also, intersecting layers should be avoided because in the image above, Core Animation needs to do a lot of work. So just to draw the red layer intersecting only the blue layer, Core Animation needs to

- cut the red layer into two pieces
- render back part of the red layer
- then render the full blue layer
- then render the front part of the red layer again

And all that work is for just intersection, and here we have multiple.

2. Building the Cube in 3D

2.1. Setting up the multi-platform project

I wanted the project to run on the macOS, iOS and iPadOS, so I used [AllApples](#) Swift package. After removing the storyboards and pimping up the AppDelegate and main.swift for the macOS version, and SceneDelegate for the mobile versions, I was ready to start.

2.1.1. main.swift for the macOS

```
import Cocoa
let delegate = AppDelegate()
NSApplication.shared.delegate = delegate
_ = NSApplicationMain(CommandLine.argc, CommandLine.unsafeArgv)
```

2.1.2. AppDelegate for the macOS

```
import Cocoa
import AllApples

class AppDelegate: NSObject, NSApplicationDelegate {
    private var window: NSWindow?
    func applicationDidFinishLaunching(_ aNotification: Notification) {
        window = AppSceneDelegate.makeWindow_Mac(theVC: CommonViewController())
    }
}
```

2.1.3. SceneDelegate for the iOS and iPadOS

```
import UIKit
import AllApples

class SceneDelegate: UIResponder, UIWindowSceneDelegate {
    var window: UIWindow?
    func scene(_ scene: UIScene, willConnectTo session: UISceneSession, options connectionOptions: UIScene.ConnectionOptions) {
        guard let aScene = (scene as? UIWindowScene) else { return }
        window = AppSceneDelegate.makeWindow_iOS(theScene: aScene, theVC: CommonViewController())
    }
}
```

2.2. Building the Basic Blocks

The first I did is make a PlainLayerView . It is an AView descendant, which means that it is typedef-ed to be either a UIView or a NSView.

It is an intermediary object just to set up a thing on the macOS as well (doesn't work yet).

I then created CustomLayerView to have nice sides for the cube, with CATextLayer as the

number of the cube side, and a nice rounded border, so that we can peek into our cube while rotating.

Here's the image of all six layers drawn by using a CustomLayerView

[Image: /images/cacube/07_original_abandoned_cube_layout.png]

This layout was abandoned because I couldn't make the transformation of purple view to work when transforming in 3D.

[Image: /images/cacube/08_side4_broken.png]

The solution is to add an additional CATransformLayer to the green layer, and mount the purple layer onto it. As explained in this blog post by Oliver Drobnik [Cubed CoreAnimation Conundrum](#).

But I didn't want to lose the linearity of the solution, and I used the approach demonstrated in the mentioned WWDC session: "2011–421 Core Animation Essentials"

They used the `zOrder` property of a layer, and so I put purple layer on top of the red layer to achieve that.

```
if number == 4 {
    view.layer.zPosition = 1
}
```

As you can see in the image below, the purple layer is in front of the red layer, which is obvious when we rotate the flattened cube.

[Image: /images/cacube/09_layers_zPosition.png]

2.3. Turning the Transform On and Off

I did take the approach that Oliver Drobnik did in his article: [Cubed CoreAnimation Conundrum](#), and used it like this:

```
side4.layer?.zPosition = on ? CACube3DView.sideWidth : 1
side1.layer?.transform = on ? CATransform3D.transformFor3DCubeSide(1,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
side2.layer?.transform = on ? CATransform3D.transformFor3DCubeSide(2,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
side3.layer?.transform = on ? CATransform3D.transformFor3DCubeSide(3,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
side4.layer?.transform = on ? CATransform3D.transformFor3DCubeSide(4,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
side5.layer?.transform = on ? CATransform3D.transformFor3DCubeSide(5,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
side6.layer?.transform = on ? CATransform3D.transformFor3DCubeSide(6,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
```

2.4. Transforming the Layers to Form a Cube

I didn't use his transform code, since he used that additional CATransformLayer, so it wouldn't work. So, here's a small extension on the CATransform3D

```

public extension CATransform3D {
    static func transformFor3DCubeSide(_ number: Int, zWidth: CGFloat) ->
    CATransform3D {

        let halfPi = CGFloat(Double.pi) / 2.0
        var trans = CATransform3DIdentity
        switch number {
            case 1:
                trans = CATransform3DMakeRotation(halfPi, 0, 1, 0)
                break
            case 2:
                trans = CATransform3DIdentity
                break
            case 3:
                trans = CATransform3DMakeRotation(-halfPi, 0, 1, 0)
                break
            case 4:
                trans = CATransform3DMakeTranslation(0, 0, zWidth)
                break
            case 5:
                trans = CATransform3DMakeRotation(-halfPi, 1, 0, 0)
                break
            case 6:
                trans = CATransform3DMakeRotation(halfPi, 1, 0, 0)
                break
            default:
                break
        }
        return trans
    }
}

```

I actually used the approach from that WWDC session, and also used `anchorPoint` properties of the `CALayer`.

2.5. Positioning Cube Sides

Here's a little extension on `CGPoint` that returns a tuple of our cube side positions and anchor points:

```

public extension CGPoint {
    static func anchorPointAndPositionForCubeSideLayer(number: Int, sideSize:
    CGFloat) -> (anchorPoint: CGPoint, position: CGPoint) {
        var resultAnchorPoint = CGPoint(x:0.5, y:0.5)
        var resultPosition = CGPoint(x:0.0, y:0.0)
        let halfSideSize: CGFloat = sideSize / 2.0
        switch number {
            case 1:
                resultAnchorPoint = CGPoint(x:1.0, y:0.5)
                resultPosition = CGPoint(x:-halfSideSize, y:0.0)
                break

```

```

        case 2:
            resultAnchorPoint = CGPoint(x:0.5, y:0.5)
            resultPosition = CGPoint(x:0.0, y:0.0)
            break
        case 3:
            resultAnchorPoint = CGPoint(x:0.0, y:0.5)
            resultPosition = CGPoint(x:halfSideSize, y:0.0)
            break
        case 4:
            resultAnchorPoint = CGPoint(x:0.5, y:0.5)
            resultPosition = CGPoint(x:0.0, y:0.0)
            break
        case 5:
            resultAnchorPoint = CGPoint(x:0.5, y:1.0)
            resultPosition = CGPoint(x:0.0, y:-halfSideSize)
            break
        case 6:
            resultAnchorPoint = CGPoint(x:0.5, y:0.0)
            resultPosition = CGPoint(x:0.0, y:halfSideSize)
            break
        default:
            break
    }
    return (anchorPoint: resultAnchorPoint, position: resultPosition)
}
}

```

In the image below I have marked where the anchor points are for each layer:

[Image: /images/cacube/10_anchor_points.png]

The only fallacy in the image above, is that the purple layer is actually above our red layer, but I wanted to show where those anchor points are. So, the actual image looks like this, but we now don't see the red layer. *[Image: /images/cacube/11_anchor_points.png]*

An Anchor point is a center of rotation. It determines how will the layer rotate. Imagine holding a playing card with two fingers. Then try to spin the card. The anchor point of that card is where you are holding it with fingers.

3. Responding to Gestures

I made a small GestureRecognizerView to be able to respond to gestures and move, rotate and scale our layers.

It hooks-up:

```

    NSPanGestureRecognizer and UIPanGestureRecognizer
    func rotate(with event: NSEvent) and UIRotationGestureRecognizer
    NSClickGestureRecognizer and UITapGestureRecognizer
    func magnify(with event: NSEvent) and UIPinchGestureRecognizer

```

It then exposes all those events to the developer to use:

```

public extension GestureRecognizerView {

```

```

@objc func rotationChanged(degrees: Float) {}
@objc func rotationChanged(radians: Float) {}
@objc func displacementChanged(displacement: CGPoint) {}
@objc func scaleChanged(scale: CGFloat) {}
@objc func tapHappened() {}
}

```

4. Building a Layer to hold a Cube

CACube3DView will hold the six layers than (can) make a cube. In order for Core Animation to render the transformed views in perspective, there is a property `subLayerTransform`.

You either use that property of your parent layer, or add another layer class to your layer hierarchy: `CATransformLayer`. I opted to use that.

```

private(set) public lazy var transformedLayer: CALayer = {
    let l = CATransformLayer()
    l.name = "Transform Layer"
    #if os(OSX)
    l.isGeometryFlipped = true
    #endif
    return l
}()

```

When adding sublayers, I add them to this `transformedLayer` property, and not my view's layer.

```

func addSideSubview(_ subview: AView) {
    addSubview(subview)

    #if os(iOS) || os(tvOS)
    transformedLayer.addSublayer(subview.layer)
    #endif

    #if os(OSX)
    if let aLayer = subview.layer {
        transformedLayer.addSublayer(aLayer)
    } else {
        fatalError("`subview.layer` == `nil`")
    }
    #endif
}

```

5. Perspective & Rotation

When the app first runs it shows in perspective. I made a little extension:

```
public extension CATransform3D {
  static func somePerspectiveTransform() -> CATransform3D {
    var perspective = CATransform3DIdentity;
    perspective.m34 = -1.0 / 500.0;
    perspective = CATransform3DRotate(perspective, CGFloat(Double.pi) / 8, 1,
0, 0);
    perspective = CATransform3DRotate(perspective, CGFloat(Double.pi) / 8, 0,
1, 0);
    perspective = CATransform3DScale(perspective, 0.7, 0.7, 0.7)
    return perspective
  }
}
```

The part: `perspective.m34 = -1.0 / 500.0;` sets the perspective. The `.34` field of a matrix shows the amount of perspective distortion applied. The amount 500 is often used in examples. If it were smaller, the layers would seem very close and distorted, like a fish-eye effect.

This is the initial transform, but we want to be able to move and rotate our cube (flattened or not) with our fingers.

Here's the code:

```
public extension CATransform3D {

  func rotationFromDisplacement(_ displacement: CGPoint, sideWidth: CGFloat,
is3D: Bool) -> CATransform3D {

    var currentTransform = self

    let totalRotation: CGFloat = sqrt(displacement.x * displacement.x +
displacement.y * displacement.y)
    let angle: CGFloat = totalRotation * .pi / 180.0

    let xRotationFactor = displacement.x / angle
    let yRotationFactor = displacement.y / angle

    if is3D {
      currentTransform = CATransform3DTranslate(currentTransform, 0, 0,
sideWidth / 2.0)
    }

    var rotationalTransform = CATransform3DRotate(currentTransform, angle,
(xRotationFactor *
currentTransform.m12 - yRotationFactor * currentTransform.m11),
(xRotationFactor *
currentTransform.m22 - yRotationFactor * currentTransform.m21),
(xRotationFactor *
currentTransform.m32 - yRotationFactor * currentTransform.m31))
  }
}
```

```

        if (is3D) {
            rotationalTransform = CATransform3DTranslate(rotationalTransform, 0, 0,
-sideWidth / 2.0);
        }

        return rotationalTransform
    }
}

```

We call it from our pan-gesture methods

```

override func displacementChanged(displacement: CGPoint) {
    guard !(displacement.x == 0 && displacement.y == 0) else { return }

    let rotationTransform =
transformedLayer.sublayerTransform.rotationFromDisplacement(displacement,
sideWidth: CACube3DView.sideWidth, is3D: isOn)
    transformedLayer.sublayerTransform = rotationTransform
}

```

We hooked up the pan-gestures prior:

```

#if os(OSX)
private func setupPanGestureRecognizer() {
    let panGR = NSPanGestureRecognizer(target: self, action:
#selector(handlePanGesture(_:)))
    addGestureRecognizer(panGR)
}
@objc func handlePanGesture(_ gestureRecognizer: NSPanGestureRecognizer) {
    let displacement: CGPoint = gestureRecognizer.translation(in: self)
    handlePan(displacement: displacement, changed: gestureRecognizer.state
== .changed)
}
#endif

#if os(iOS) || os(tvOS)
private func setupPanGestureRecognizer() {
    let panGR = UIPanGestureRecognizer(target: self, action:
#selector(handlePanGesture(_:)))
    addGestureRecognizer(panGR)
}
@objc func handlePanGesture(_ gestureRecognizer: UIPanGestureRecognizer) {
    let displacement: CGPoint = gestureRecognizer.translation(in: self)

    handlePan(displacement: displacement, changed: gestureRecognizer.state
== .changed)

    if gestureRecognizer.state == .changed {
        gestureRecognizer.setTranslation(.zero, in: self)
    }
}
}

```



```
#endif
```

We can also add a simple rotation transform for the rotation events/ gestures.

```
override func rotationChanged(radians: Float) {
    let transform = transformedLayer.sublayerTransform
    let rot = CATransform3DRotate(transform, CGFloat(radians), 0, 1, 0)
    transformedLayer.sublayerTransform = rot
}
```

And scale, in all three axes:

```
override func scaleChanged(scale: CGFloat) {
    let scaleTransform = CATransform3DScale(transformedLayer.sublayerTransform,
scale, scale, scale)
    transformedLayer.sublayerTransform = scaleTransform
}
```

The tap turns on and off our 3D transform

```
override func tapHappened() {
    set3DCube(on: isOn)
}
```

Here's the 3D cube code

```
func set3DCube(on: Bool) {
    side4.layer.zPosition = on ? CACube3DView.sideWidth : 1
    side1.layer.transform = on ? CATransform3D.transformFor3DCubeSide(1,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
    side2.layer.transform = on ? CATransform3D.transformFor3DCubeSide(2,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
    side3.layer.transform = on ? CATransform3D.transformFor3DCubeSide(3,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
    side4.layer.transform = on ? CATransform3D.transformFor3DCubeSide(4,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
    side5.layer.transform = on ? CATransform3D.transformFor3DCubeSide(5,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
    side6.layer.transform = on ? CATransform3D.transformFor3DCubeSide(6,
zWidth: CACube3DView.sideWidth) : CATransform3DIdentity
}
```

We either set the identity transform, which means no transform, to our cube sides, or the transform appropriate for that particular side.

6. macOS Troubles

I suppose I need to do further investigation in how macOS treats layers of `NSView`, for this little experiment is not working on the macOS, yet. Here's how the flattened cube looks on the macOS

[Image: /images/cacube/12_cube_macOS_flat_messed_up.png]

So, the positioning of the layers is not respected.

And here is the cube in 3D:

[Image: /images/cacube/13_cube_macOS_3D_messed_up.png]

I did try to force the `isGeometryFlipped = true` everywhere. Anyway this needs more work.

If you want to help with the macOS implementation, please, be my guest.

Here's a link to a GitHub repo with the source code: [Core Animation 3D Cube](#)